

Campaign roadmap: staged predictive modelling of PEGDA photopolymerization

Lab note for the Photopolymerization campaign
Companion to the chemistry/process background note

13 September 2026

Abstract

This note is the *operational* roadmap. The companion background note (`docs/background_photopolymerization.p`) defines the chemical objects. Here we define the *sequence of scientific questions*, what is done at each stage, what counts as success, which gates must pass before the next stage is opened, and how reasonability and literature agreement are judged without confusing a plot that “looks like polymerization” with a result that may be cited. No stage below has been executed in this repository as of this date: every row is **proposed**.

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1 Purpose and locked scientific object

Object. Predict acrylate conversion $p(\mathbf{x}, t)$ of a PEGDA chain-growth photopolymerization from a known illumination history, and only then obtain gelation, loop-aware elastically active density η , and shear modulus G [Zhu et al., 2020]. Later stages add a projected surface irradiance [Meenakshisundaram et al., 2020, Montgomery et al., 2022] and a prescribed plug flow [Slutzky et al., 2019].

Why stages exist. The five campaign papers close *different layers* of the same process (kinetics, optics, transport, network). A single three-dimensional calculation that mixes all of them can run and still not tell you which layer is wrong. Each stage therefore owns one claim, one literature counterpart, and one set of gates.

Locked non-goals (this ladder). The full multi-species model of Dobson and Bowman [2024] is a *reference*, not the everyday constitutive law. Navier–Stokes coupled to gelation, scattering optics, inverse grayscale design, PNIPAM, and thiol–ene chemistry are parked. Eosin Y bleaching kinetics [Zhu et al., 2020] must not be dropped into an Irgacure Type I rate law without an explicit chemistry flag. A Jacobs working-curve depth [Jacobs, 1992] may be plotted as a diagnostic overlay; it is not the polymer state.

Reconciliation with the broader Undermind notes. A longer literature synthesis exists in the DFG modelling workspace. That synthesis is right about the coupled process (history-dependent kinetics, oxygen transport, independent pattern and resin velocities). It over-specifies the *first* calculation if read as “implement Dobson in two dimensions with a measured point-spread

function.” This roadmap follows the Undermind *Stage 1 PEGDA* notes: a local intensity-driven constitutive response first; Dobson used to stress-test closures; measured projection and flow added only after that response is internally sound.

2 How a stage is run (process, not software brands)

Every stage produces the same *scientific quartet*:

1. a frozen parameter/configuration record for *this claim only*;
2. a forward calculation and an overview figure of the quantities the claim is about;
3. a validator that prints pass/fail per named gate and an overall line;
4. a journal paragraph with exactly one evidence tag.

Then **stop**. Do not open the next stage because the previous run finished. An ensemble that is finite and free of NaN is *engineering success*. It is not scientific validity.

2.1 Evidence tags

Table 1: Mandatory primary tag for each stage. Never upgrade a prior stage without re-running its validator.

Tag	Meaning
proposed	Designed, not run. This document.
smoke-tested	Short run; phenomenology only (signs, induction, no crash).
validated	Named internal gates pass for the <i>stated claim</i> of that stage.
matched	Numbers compared to a named experiment or paper figure under the <i>same formulation</i> . Rare.
invalid	Runnable, but must not be cited (wrong physics, mixed recipes, superseded). Keep files; quarantine in the campaign index.

A pretty conversion curve never upgrades **smoke-tested** to **validated**. Matching Montgomery’s FTIR $p(t)$ never upgrades Zhu’s modulus, because those are different chemistries.

2.2 Three kinds of “agreement with literature”

Confusion here is the usual way a campaign becomes un-citable. Distinguish:

Phenomenological agreement The *sign* and *ordering* of effects match a paper: oxygen delays conversion; equal dose at higher intensity need not give the same p [Wydra et al., 2014, Lin et al., 2019]; a PDMS wall thickens an inhibition layer [Dendukuri et al., 2008]; dilute PEGDA may never gel [Zhu et al., 2020]. This can support **smoke-tested** or part of **validated**. It does *not* license transferring fitted numbers.

Analytic identity In a reduced limit that the paper actually solved, the calculation reproduces an algebraic relation to a stated tolerance (for example Slutzky’s $C_O/C_{O,0} =$

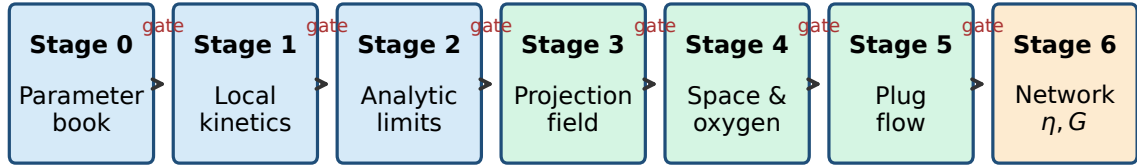
$(C_M/C_{M,0})^{k_O/k_P}$ when k_P and k_O are constant). This is the strongest *internal* literature test and is required for **validated** on Stage 2.

Quantitative match A number from a named figure or table is recovered within a pre-declared band, using *that paper's* recipe, initiator, wavelength, and geometry. Only then may the tag become **matched**. Montgomery's $k_P(p)$ for PEGDA 250/Irgacure 819 does not **matched** Zhu's PEGDA 575/eosin Y hydrogel.

Reasonability (orders of magnitude). Before any match is claimed, ask: are concentrations in mol m^{-3} (or a declared unit system) consistent with the formulation mass fractions? Is photolysis βI a frequency of order 10^{-3} – 10^{-1} s^{-1} under the paper's I , not 10^6 because milliwatts were treated as watts [Montgomery et al., 2022]? Is D_{O_2} of order $10^{-10} \text{ m}^2 \text{ s}^{-1}$ in a liquid, not 10^{-5} ? Is P_{gel} between 0 and 1? A calculation that violates these is **invalid** even if gates on positivity nominally pass by clipping.

3 The ladder at a glance

Campaign ladder (stop at each gate; do not skip)



Parked after Stage 6: moving pattern, free-volume Dobson chemistry, heat, inverse design.

Figure 1: Seven active stages. Each arrow is a gate: the next box is not opened on engineering success alone. After Stage 6, pattern motion, Dobson free-volume chemistry, heat, and inverse design remain parked.

Dimensionless groups decide *which* spatial physics to turn on, not whether the previous stage passed:

Group	Meaning	If $\mathcal{O}(1)$ or large
$\beta_{\text{opt}} = AH$	optical depth	resolve z (Stage 4)
Da_O	oxygen reaction vs diffusion	wall dead-zone [Dendukuri et al., 2008]
Π_{flow}	photolysis vs advected O_2	gel/no-gel in flow [Slutzky et al., 2019]
$\tau_{\text{exp}} = t_{\text{uv}}/t_L$	pulse vs residence	fiber-length collapse
$\Lambda_v = \mathbf{v}_{\text{pattern}} - \mathbf{u} /U_*$	relative pattern motion	do not identify scan with flow
$\text{Pe}_i = UH/D_i$	advection vs diffusion	which species need spatial transport
$\chi = \Delta t_{\text{frame}}/t_{\text{chem}}$	frame vs chemistry	keep discrete DMD frames

Table 2: One-line claim per stage and the literature that owns the claim.

Stage	Claim in one line	Literature owner
0	Every coefficient has a provenance or an explicit “measure” flag.	Tables in Montgomery et al. [2022], Slutzky et al. [2019], Zhu et al. [2020], Dobson and Bowman [2024]
1	Prescribed $I(t)$ maps to $p(t)$ with oxygen induction and history dependence.	Montgomery et al. [2022]; reciprocity [Wydra et al., 2014, Lin et al., 2019]
2	Reduced limits recover algebraic identities, not only curves.	Slutzky et al. [2019] Eq. 3; QSS radicals [Zhu et al., 2020]; R_p-I exponent [Dobson and Bowman, 2024]
3	Bitmap, kernel, and scan produce a conserved surface irradiance.	Meenakshisundaram et al. [2020], Montgomery et al. [2022]
4	The same kinetics in space show a wall dead-zone and grayscale blur.	Montgomery et al. [2022], Dendukuri et al. [2008]
5	Plug flow recovers a critical speed and t_{uv}/t_L organisation.	Slutzky et al. [2019]
6	p and $[PEGDA]_0$ yield P_{gel} , $\eta \leq 1$, G below ideal.	Zhu et al. [2020], Macosko and Miller [1976]

4 Cross-cutting reflection protocol

At the end of *every* stage, before the tag is written, answer these five questions in the journal. If any answer is “no” or “not applicable because we mixed recipes,” the tag cannot be **matched**, and **validated** is in doubt.

1. **Which predicate did we compute?** Dose, surface I , in-resin I , p , a 2% double-bond threshold [Slutzky et al., 2019], Macosko P_{gel} [Zhu et al., 2020], η , or G ? The background note lists the forbidden equivalences; repeating one of them is an automatic **invalid**.
2. **Which paper actually owns this layer?** Cite equation or table numbers. If the owner is Montgomery and the numbers came from Zhu, the comparison is phenomenological at best.
3. **What would make the result unreasonable even if the solver is green?** Wrong units of I and β ; $p > 1$; gelation at $p = 0$ for a dilute precursor that Zhu predicts cannot gel; a “grayscale” edge sharper than the optical kernel while diffusion is on; a flow case that gels at infinite U .
4. **Did equal-dose protocols collapse?** If they *do* collapse for a Type I acrylate with oxygen and bimolecular termination, the constitutive law is suspect [Wydra et al., 2014] unless oxygen is absent and k_t is first-order, which must be stated.
5. **What is still required?** Name the measurement (FTIR $p(t)$, radiometer PSF, oxygen

probe, rheology G) that would be needed to move **validated** to **matched**.

Fit discipline. Parameters are identified in this order, not all at once [Montgomery et al., 2022, Dobson and Bowman, 2024]: (i) photolysis $\beta C_{I,0}$ from initiator decay or very-low conversion; (ii) k_O and $C_{O,0}$ from induction time or inhibition thickness; (iii) k_{p0} and termination pieces from *several* intensities, including equal-dose pairs; (iv) conversion sensitivity c_p from high-conversion slowdown; (v) network C_0 from gelation versus concentration, independently of the rate fit [Zhu et al., 2020]. A single conversion curve used to fit ten numbers is **invalid** as a parameter inference even if the forward gates pass.

5 Stage 0 — Parameter book

5.1 Scientific question

Can every coefficient that will enter later stages be traced to a paper table, a recipe calculation, or an explicit gap labelled **still required**?

5.2 What you do

Transcribe, with units and initiator class:

- Montgomery Table 1 (optical attenuation, pixel Gaussian fallback) and the $k_p(p)$, $k_t(p)$ closures [Montgomery et al., 2022];
- Slutzky Table 2 and the stated k_d window [Slutzky et al., 2019];
- Zhu’s C_0 , functionality $f = 2$, and the statement that PEGDA 575 is the hydrogel example [Zhu et al., 2020];
- Dobson’s role as HDDA reference, not as a PEGDA number dump [Dobson and Bowman, 2024].

Flag molecular-weight mismatch (250 vs 575 g mol^{−1}) and Type I vs Type II in the same table. Do not fit anything yet.

5.3 Success factors

Provenance is complete; silent defaults are forbidden; SI (or a declared conversion to Montgomery’s intensity convention) is written next to β .

5.4 Stage gates

- Every parameter has a source column: paper / calculated from recipe / **still required**.
- No row mixes PEGDA 250 dry film, PEGDA 575 water gel, and HDDA without a chemistry key.
- Intensity units for β are stated.

5.5 Reasonability and literature

Reasonable if mass-fraction $\rightarrow$ molarity conversions recover $[M]_0$ of order several mol L⁻¹ for neat diacrylate and lower for diluted hydrogels, consistent with the papers' recipes. Unreasonable if one "PEGDA" row is used for all three formulations.

5.6 Tag and stop

proposed until the book exists; then validated for the *bookkeeping* claim only. Stop. Do not solve ODEs in this stage beyond a unit-smoke if needed to read the file.

6 Stage 1 — Local constitutive response

6.1 Scientific question

For a *prescribed* $I(t)$, with no space and no flow, do initiator, radicals, oxygen, and acrylate evolve as a stiff free-radical system with conversion-dependent k_p and k_t [Montgomery et al., 2022]?

The balances (Montgomery Eqs. 16–19 with diffusion switched off) are

$$\dot{C}_I = -\beta I C_I, \tag{1}$$

$$\dot{C}_R = m\beta I C_I - 2k_t(p)C_R^2 - k_O C_O C_R, \tag{2}$$

$$\dot{C}_O = -k_O C_O C_R, \tag{3}$$

$$\dot{C}_M = -k_p(p) C_R C_M, \quad p = 1 - C_M/C_{M,0}. \tag{4}$$

Temperature is held fixed. Radicals are *not* replaced by a quasi-steady algebraic law at this stage: the stiffness is physical [Montgomery et al., 2022, Dobson and Bowman, 2024].

6.2 What you do

Integrate (1)–(4) for a protocol family: dark; constant I ; square pulse; two equal-dose pairs (I_1, t_1) and (I_2, t_2) with $I_1 t_1 = I_2 t_2$; oxygen-free limit $C_{O,0} = 0$. Export $p(t)$, $C_O(t)$, $R_p(t)$. Thermal energy stays off.

6.3 Success factors

- Dark control is dead: no photoinitiator-driven conversion.
- An oxygenated sample shows an induction period; the oxygen-free twin does not [Decker and Jenkins, 1985, Dendukuri et al., 2008].
- Equal-dose pairs do *not* collapse in general [Wydra et al., 2014, Lin et al., 2019].
- High conversion slows as $k_p(p)$ and $k_t(p)$ intend [Montgomery et al., 2022, Anseth et al., 1994].

- Concentrations stay non-negative; $0 \leq p \leq 1$; C_I and C_M decrease under illumination.

6.4 Stage gates (all required for validated)

Gate	Pass criterion
Dark	$I = 0 \Rightarrow \Delta p$ below solver tolerance.
Positivity	$C_i \geq -10^{-12} \text{ mol m}^{-3}$ without ad-hoc clipping of large negatives.
Bounds	$0 \leq p \leq 1$.
Monotone PI	C_I nonincreasing when $I > 0$.
Acrylate balance	$\int R_p dt$ matches ΔC_M to a stated relative tolerance.
Reciprocity	At least one equal-dose pair with $ p_1 - p_2 $ at a common dose <i>not</i> both consistent with numerical noise.
Induction	With $C_{O,0} > 0$, p remains near 0 until C_O has dropped; with $C_{O,0} = 0$, that delay vanishes.

6.5 Reasonability and literature

Compare *shapes* to Montgomery’s FTIR conversion-versus-time family [Montgomery et al., 2022] and to Wydra’s dose plots [Wydra et al., 2014]. Do *not* declare **matched** unless the formulation (PEGDA 250, 0.1 wt% Irgacure 819, their I in their units) is the one in the file. If equal-dose curves collapse, first check that k_t was accidentally made first-order or that oxygen was omitted; that is a constitutive bug, not a discovery.

Quasi-steady radicals (5) are a *diagnostic overlay* in this stage, not a substitute for (2):

$$C_R \overset{?}{\approx} \sqrt{\frac{m\beta IC_I}{2k_t}} \quad (C_O = 0, \text{ slow transients}). \quad (5)$$

6.6 Tag and stop

smoke-tested after a short protocol set; **validated** after the full gate table. **matched** only against a named Montgomery (or in-house FTIR) series. Stop. Do not add diffusion here.

7 Stage 2 — Analytic and reduced limits

7.1 Scientific question

Does the constitutive law contain the limits the literature actually solved, or only a plausible S-shaped $p(t)$?

7.2 What you do

Introduce a *constant-coefficient fork* of (1)–(4) (constant k_p , k_t , k_O ; still no space). Test:

- Slutzky’s algebraic oxygen–monomer relation

$$\frac{C_O}{C_{O,0}} = \left(\frac{C_M}{C_{M,0}} \right)^{k_O/k_P} \quad (6)$$

which follows from dividing the M and O_2 balances when those two rates dominate [Slutzky et al., 2019];

- oxygen-free QSS radicals against (5) after the fast transient [Zhu et al., 2020];
- qualitative R_p versus I : exponent near $1/2$ when bimolecular termination wins, nearer 1 when oxygen makes radical loss pseudo-first-order [Dobson and Bowman, 2024].

The last test is *not* a demand to reproduce Dobson’s HDDA numbers.

7.3 Success factors

Identities hold to a pre-declared `rtol/atol`. The R_p – I exponent moves in the direction Dobson describes when $C_{O,0}$ is raised at fixed low I .

7.4 Stage gates

- Equation (6) on the constant-rate fork: maximum relative residual below the declared band (e.g. 10^{-4} away from $C_M \rightarrow 0$).
- QSS residual $|C_R - \sqrt{R_i/2k_t}|/C_R$ small after a stated number of radical times, for $C_{O,0} = 0$.
- Full conversion-dependent model is *not* required to satisfy (6); the journal must say so, or the test is meaningless.

7.5 Reasonability and literature

This is the campaign’s manufactured-solution analogue: we test the algebra the papers derived, not their figures. Failure here means the right-hand side was transcribed wrongly. Do not “fix” k_t to make (6) hold in the conversion-dependent model.

7.6 Tag and stop

`validated` if identities pass. No `matched` is available at this stage (no experiment yet). Stop.

8 Stage 3 — Projection engine

8.1 Scientific question

Given a bitmap, a spatial kernel, and optional scan motion, what is the surface irradiance $I_\perp(x, y, t)$, and is energy accounted for [Meenakshisundaram et al., 2020, Montgomery et al., 2022]?

This stage has *no chemistry*.

8.2 What you do

- Implement superposition of pixel kernels. Montgomery’s Gaussian (their Eq. 7) is the fallback [Montgomery et al., 2022].
- Allow a measured, non-Gaussian kernel as the scientifically preferred input [Meenakshisundaram et al., 2020].
- Accumulate dose $E_{\perp} = \int I_{\perp} dt$ under a scan whose frame period is projected pitch divided by scan speed [Meenakshisundaram et al., 2020].
- Optionally overlay a Jacobs $C_d = D_p \ln(E/E_c)$ field, labelled **diagnostic, not state** [Jacobs, 1992].

8.3 Success factors

A single pixel’s integrated power matches the commanded pixel power. Overlapping Gaussians produce the qualitative bleed of Montgomery’s Fig. 3. Scan: if dwell $\propto 1/v$, peak $E_{\perp}$ is approximately v -independent; if not, the journal explains why (frame discreteness, χ not small).

8.4 Stage gates

- Power conservation of one pixel to a stated relative error.
- Ten-pixel overlap: irradiance at a commanded edge is not a top-hat (bleed exists).
- Dark bitmap $\Rightarrow I_{\perp} = 0$.
- Chemistry disabled: no p is reported as a Stage 3 result.

8.5 Reasonability and literature

Compare kernel shape to Meenakshisundaram’s measured profiles (non-Gaussian, possible flat top). If a Gaussian is used, the journal must say the projection error relative to a measured PSF is **still required**. Do not compare printed widths from Jacobs overlays to Montgomery conversion fields and call it validation.

8.6 Tag and stop

validated on conservation and bleed. **matched** only if a camera/radiometer map of *this* projector is used. Stop. Do not couple attenuation yet.

9 Stage 4 — Spatial reaction–diffusion and light

9.1 Scientific question

If the Stage 1 right-hand side is evaluated in space, with Beer–Lambert attenuation and conversion-dependent diffusivities, do we obtain (i) depth structure and (ii) lateral blur beyond the optical kernel [Montgomery et al., 2022]? As a variant: does an oxygen Dirichlet or permeable wall create a dead zone [Dendukuri et al., 2008]?

The optical reduction is

$$\frac{\partial I}{\partial z} + A(\mathbf{x}, t)I = 0, \quad A = \alpha_I C_I + pA_{\text{poly}} + (1 - p)A_{\text{mon}} + A_{\text{abs}}, \quad (7)$$

following Montgomery et al. [2022]. Species follow

$$\partial_t C_i = \nabla \cdot (D_i(p) \nabla C_i) + R_i, \quad (8)$$

with the same R_i as Stage 1. First geometry: one-dimensional z . Second: an x – z grayscale strip. Not three dimensions.

Two boundary-condition variants must be labelled:

Sealed slides no-flux, Montgomery’s FTIR cell.

Oxygen wall fixed C_O or permeable flux, Dendukuri/CLIP-type dead zone [Dendukuri et al., 2008, Tumbleston et al., 2015].

They are different experiments.

9.2 What you do

Method of lines in space; local stiff reaction as in Stage 1. Harmonic liquid–solid $D_i(p)$ as in Montgomery, unless a later stage replaces it by Dobson free volume (parked). Run: dark box; uniform illumination, sealed; uniform illumination, oxygen wall; grayscale step $G_0|G_{80}$.

9.3 Success factors

- Dark sealed box conserves $\int C_M$.
- Optically thick $\beta_{\text{opt}} \gtrsim 1$ produces a conversion front, not a uniform $p(z)$ [Cabral and Douglas, 2005, Zhu et al., 2020, Montgomery et al., 2022].
- Oxygen wall: a layer of near-zero p at the boundary; sealed slides: much weaker surface inhibition.
- Grayscale: chemical transition width $>$ optical σ when radical or monomer diffusion is on [Montgomery et al., 2022].

9.4 Stage gates

- Conservation (sealed, dark) to a stated mass residual.
- Manufactured diffusion-only problem (no reaction) recovers a known Gaussian spread or similar.
- Fourier/CFL (or implicit stability) documented; no grid-scale ringing in C_R that changes p by more than the gate band under refinement.
- Grayscale width test as above, with diffusion on versus off.
- The Stage 1 ODE at one collocation point with the local $I(t)$ extracted from the field agrees with the spatial cell (same RHS).

9.5 Reasonability and literature

Phenomenology: Montgomery’s overcure and edge blur; Dendukuri’s inhibition layer. **matched** requires their geometry, absorber loading, and PEGDA 250 recipe, or an in-house film with measured $I(z = 0)$. Unreasonable: a dead zone in a sealed, deoxygenated film; or no dead zone against air/PDMS when Da_O is large.

9.6 Tag and stop

validated on conservation, MMS, and the two qualitative structures (front, blur or dead zone). Stop. Do not add flow.

10 Stage 5 — Plug-flow benchmark

10.1 Scientific question

If material is carried at uniform speed U through a fixed illumination window of length L , do initiator, radicals, monomer, and oxygen follow Slutzky’s steady balances, and does a critical speed separate gel from no-gel [Slutzky et al., 2019]?

$$U \frac{dC_i}{dx} = R_i(C, I(x)), \quad I(x) > 0 \text{ only on } [0, L]. \quad (9)$$

This is the Lagrangian twin of Stage 1 under $t = x/U$ if diffusion is negligible, as Slutzky argue for their jet ($\sqrt{Dt} \ll w_1$).

Use *their* operational gel predicate $C_M/C_{M,0} = 0.98$ when comparing to their fiber onset—and write that sentence explicitly, because it is not Zhu’s P_{gel} [Zhu et al., 2020, Slutzky et al., 2019].

10.2 What you do

Integrate in x with their Table 2 as a labelled parameter set. Sweep U and I . Form $\tau_{\text{exp}} = t_{\text{uv}}/t_L$ for pulsed cases. Do not enable wall oxygen diffusion unless a variant is named “not Slutzky’s

jet.”

10.3 Success factors

- $U \rightarrow \infty$: negligible conversion in the window.
- Small U , long t_{uv} : the 2% threshold is crossed.
- U_c increases with I and with $[PI]_0/[O_2]_0$ in the sense of their scaling.
- Fiber-length organisation against t_{uv}/t_L is monotonic in the uniform-fiber regime they describe.

10.4 Stage gates

- The two limits (fast flow / slow flow) as above.
- Constant-rate fork still satisfies (6) along x .
- Gel predicate documented: 2% vs Macosko, never mixed in one figure without a dual axis and a caption warning.

10.5 Reasonability and literature

`matched` is allowed only against Slutzky Table 2 and their figures, with PEGDA 575 jet chemistry—not Montgomery thin-film rates. Unreasonable: gelation that is independent of $[O_2]_0$ while k_O is large; or a critical speed that decreases as I increases.

10.6 Tag and stop

Qualitative scaling: `validated`. Their numbers: `matched` if within a pre-declared band (they report tight length statistics; do not demand 2.9% dimensional error from a different paper). Stop.

11 Stage 6 — Network post-processor

11.1 Scientific question

Given p and $[PEGDA]_0$, when does a sample-spanning network appear, and what elastically active fraction η and phantom modulus G follow, including loops [Zhu et al., 2020, Macosko and Miller, 1976, Miller and Macosko, 1976]?

$$\theta = \frac{[PEGDA]_0}{[PEGDA]_0 + 2C_0}, \quad P_{\text{gel}} = \frac{1 - \theta}{2\theta} = \frac{C_0}{[PEGDA]_0}, \quad (10)$$

$$G = \eta N_A [PEGDA]_0 k_B T, \quad \eta \leq 1. \quad (11)$$

This stage does *not* advance C_I or C_R . It maps fields from Stages 1, 4, or 5.

11.2 What you do

Implement the recursive probabilities as in Zhu (pre- and post-gel). Sweep $[\text{PEGDA}]_0$ through C_0 . Apply $\eta(p)$ to a conversion field from an earlier stage without refitting k_p .

11.3 Success factors

- $[\text{PEGDA}]_0 \leq C_0 \Rightarrow$ no gel at $p = 1$.
- $\theta \rightarrow 1 \Rightarrow P_{\text{gel}} \rightarrow 0$.
- $G < G_{\text{ideal}}$ when loops are allowed; $\eta = p^2$ recovered in the no-loop limit they state.

11.4 Stage gates

The three bullets above as tests. Chemistry flag: eosin Y bleaching ODEs are off unless a Type II branch is explicitly selected. No use of G to retune Stage 1 rates.

11.5 Reasonability and literature

Compare gel-versus-concentration and G versus p to Zhu's rheology/indentation *only* for their eosin Y PEGDA 575 water system. Applying η to a Montgomery Type I conversion field is a *hypothetical* network map: allowed if labelled, not **matched** to Zhu's figures.

11.6 Tag and stop

Algebraic limits: **validated**. Zhu's mechanical data: **matched** only on their formulation. Park FEM, swelling, and inverse design.

12 Parked stages (not opened by this note)

Item	Reopen when	Needed
Moving $\mathbf{v}_{\text{pattern}} \neq 0$	Stage 3 validated and Λ_v not ≈ 0 in the experiment	scan kinematics
Eulerian–Lagrangian markers	Stages 4–5 validated	$\mathbf{u}(\mathbf{x})$
Dobson free-volume / chain-length k_t	equal-dose tests fail after a correct Stage 1 transcription	two-intensity FTIR
Energy equation	isothermal $p(t)$ systematically wrong vs measured T	$\Delta H_p, h$
Inverse DMD	Stage 4 grayscale widths matched	target η maps

13 What this campaign will refuse to cite

- Jacobs depth as conversion, gelation, or modulus.

- Montgomery coefficients as universal PEGDA constants.
- A 2% double-bond “gel” plotted as Zhu’s P_{gel} without a caption.
- Equal-dose collapse presented as a virtue of the acrylate Type I model.
- Any Stage n figure used as evidence for Stage $n + 1$ ’s claim.

14 Immediate next action

The planned Stages 0–6 ladder is complete (13 September 2026). Parked items (moving pattern, Eulerian markers, Dobson free volume, energy equation, inverse DMD) stay parked until a new request.

Stage 0 evidence. validated for the bookkeeping claim. Twenty-plus parameters (kinetics, Table 1 optics, Table 3 D_i); $C_{O,0}$ remains **still required**.

Stage 1 evidence. validated for internal constitutive gates. Not **matched** to FTIR.

Stage 2 evidence. validated for analytic identities on the *constant-rate fork*: Slutzky Eq. 3 residual 1.1×10^{-6} ; oxygen-free QSS residual 2.1×10^{-3} after 0.5 s. The conversion-dependent Montgomery RHS is *not* required to obey Eq. 3 at high p ; during the oxygen window $p \approx 0$ so Eq. 3 also holds there (logged, not a fail). Not **matched** to experiment.

Stage 3 evidence. validated for Montgomery Eq. 7 Gaussian fallback: 1-D/2-D pixel integrals match the analytic $2\sigma\sqrt{\pi}$ and $4\pi\sigma^2$ forms; dark bitmap; ten-pixel bleed; scan $\max(E)v$ independent of v . Not **matched** to a measured DMD kernel. Grayscale $I(g)$ is a linear assumption, not the SI RGB calibration. Jacobs working-curve depth was not used.

Stage 4 evidence. validated internally, not **matched** to printed geometries. Sealed dark $\int C_M$ residual 0; cosine MMS relative error 6.9×10^{-3} ; $D_i = 0$ surface cell matches Stage 1 ($|\Delta p| \sim 4 \times 10^{-6}$). Thick-vat absorber $w = 0.06$ wt%: $\beta_{\text{opt}} \approx 1.50$, $p(0) - p(H) = 0.52$ at 8 s on $H = 200 \mu\text{m}$. Oxygen Dirichlet at $z = 0$: $p_{\text{wall}} = 0.007$ vs $p_{\text{mid}} = 0.53$. G0|G80 10–90% width $\approx 53 \mu\text{m} > \sigma$; Table 3 D_i add $0.2 \mu\text{m}$ at 8 s (PSF-dominated interface). No flow.

Stage 5 evidence. validated on Slutzky Table 2 chemistry (PEGDA 575 jet), not **matched** to their measured U_c map. Fig. 2 high-oxygen illustration: $\min M/M_0 = 1.00$ (no gel). Fig. 3 low-oxygen: $\min M/M_0 = 0.863$ (crosses the operational 0.98 predicate; paper reports $M \simeq 0.86$). Eq. 3 along x : residual 7.9×10^{-7} . $U = 1 \text{ ms}^{-1}$: negligible conversion. U_c scales $\propto k_d$ and $\propto [\text{PI}]_0$. Kinematic Ut_{uv} increases with pulse length. Gel predicate is $M/M_0 = 0.98$, not Zhu P_{gel} . Montgomery rates and wall O_2 diffusion were not used.

Stage 6 evidence. validated for algebraic network identities, not matched to Zhu rheology/indentation. $P_{\text{gel}} = C_0/[\text{PEGDA}]_0$ with $C_0 = 3.45$ vol% (Table A2). $[\text{PEGDA}]_0 \leq C_0$: $\eta(p=1) = 0$. $\theta \rightarrow 1$: $P_{\text{gel}} \rightarrow 0$. No-loop: $\eta = p^2$. At 20 vol% with loops: $\eta(p=1) = 0.378 < 1$, $G/G_{\text{ideal}} = 0.378$. Eosin Y bleaching ODEs off. A Montgomery $p(t)$ mapped through η is labelled hypothetical. k_p was not refit from G . Zhu $P_{\text{gel}}(20\%) = 0.173$ is not Slutzky’s 2% threshold.

Parked: inverse DMD, swelling FEM, Type II bleaching kinetics.

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